EAPCET Chemistry — Atomic Structure

A. Discovery of Sub-atomic Particles & Early Atomic Models

1. Which experiment led to the discovery of the atomic nucleus and revealed that most of the atom is empty space?

(A) Millikan's oil drop experiment
(B) Rutherford's alpha-particle scattering experiment
(C) Thomson's cathode ray experiment
(D) Chadwick's neutron experiment

2. Thomson's plum-pudding model of the atom mainly failed to explain:

(A) the existence of electrons
(B) the overall mass of the atom
(C) the scattering of alpha particles through large angles
(D) the charge on the electron

3. In Rutherford's scattering experiment, most alpha particles passed straight through the gold foil because:

(A) the nucleus is negatively charged
(B) atoms are mostly empty space
(C) alpha particles are too heavy to be deflected
(D) gold foil is non-metallic

4. The approximate charge-to-mass ratio (e/m) of the electron, as determined by J. J. Thomson, is:

(A) 1.76×10^{11} C kg^{-1}
(B) 9.1×10^{-31} C kg^{-1}
(C) 1.6×10^{-19} C kg^{-1}
(D) 6.02×10^{23} C kg^{-1}

5. The sub-atomic particle discovered by James Chadwick, found to carry no charge, is the:

(A) proton
(B) positron
(C) neutron
(D) electron

B. Atomic Number, Mass Number, Isotopes & Isoelectronic Species

6. Two atoms having the same atomic number but different mass numbers are called:

(A) isobars
(B) isotones
(C) isotopes
(D) isoelectronic species

7. Species that possess the same number of electrons are described as:

(A) isotopes
(B) isobars
(C) isoelectronic species
(D) isotones

8. An atom of an element has mass number 40 and contains 21 neutrons. The number of electrons present in the neutral atom is:

(A) 19
(B) 21
(C) 40
(D) 61

9. Which of the following pairs represents isotones (species with the same number of neutrons)?

(A) ^{14}C and ^{14}N
(B) ^{14}C and ^{15}N
(C) ^{35}Cl and ^{37}Cl
(D) ^{40}Ar and ^{40}Ca

C. Electromagnetic Radiation & the Photoelectric Effect

10. Arrange the following electromagnetic radiations in increasing order of energy per photon: microwaves, X-rays, visible light, radio waves.

(A) radio waves < microwaves < visible light < X-rays
(B) X-rays < visible light < microwaves < radio waves
(C) visible light < radio waves < microwaves < X-rays
(D) microwaves < radio waves < X-rays < visible light

11. A photon has a wavelength of 400 nm. Its energy is approximately ($h = 6.63 \times 10^{-34}$ Js, $c = 3 \times 10^{8}$ m s^{-1}):

(A) 4.97×10^{-19} J
(B) 4.97×10^{-25} J
(C) 1.66×10^{-27} J
(D) 3.31×10^{-19} J

12. If the work function of a metal is 2.14 eV, its threshold wavelength for photoelectric emission is approximately (use hc ≈ 1240 eV nm):

(A) 580 nm
(B) 400 nm
(C) 650 nm
(D) 300 nm

13. Light of frequency 8×10^{14} Hz strikes a metal whose threshold frequency is 5×10^{14} Hz. The maximum kinetic energy of the ejected photoelectron is ($h = 6.63 \times 10^{-34}$ Js):

(A) 1.99×10^{-19} J
(B) 3.32×10^{-19} J
(C) 5.30×10^{-19} J
(D) 1.33×10^{-19} J

14. According to Einstein's photoelectric equation, increasing the intensity of incident light (frequency held above the threshold) increases:

(A) the kinetic energy of each photoelectron
(B) the number of photoelectrons emitted per second
(C) the threshold frequency of the metal
(D) the work function of the metal

15. A 60 W sodium lamp emits light of wavelength 589 nm. The number of photons it emits per second is closest to ($h = 6.63 \times 10^{-34}$ Js, $c = 3 \times 10^{8}$ m s^{-1}):

- **(A)** 1.8×10^{20}
- **(B)** 1.8×10^{23}
- **(C)** 3.6×10^{19}
- **(D)** 6.0×10^{20}

16. The wavelength associated with a radio wave of frequency 88 MHz is:

- **(A)** 3.4 m
- **(B)** 34 m
- **(C)** 0.34 m
- **(D)** 340 m

D. Bohr's Model of the Hydrogen Atom & Hydrogen-like Species

17. According to Bohr's postulate, the angular momentum of an electron in the nth orbit is quantized as:

- **(A)** $mvr = nh$
- **(B)** $mvr = nh/2\pi$
- **(C)** $mvr = n^2h/2\pi$
- **(D)** $mvr = nh^2/2\pi$

18. The radius of the first Bohr orbit of the hydrogen atom is 52.9 pm. The radius of the second orbit of the He^{+} ion is:

- **(A)** 52.9 pm
- **(B)** 105.8 pm
- **(C)** 211.6 pm
- **(D)** 26.45 pm

19. The energy of the electron in the ground state of hydrogen is −13.6 eV. Its energy in the third excited state ($n = 4$) is:

- **(A)** −3.4 eV
- **(B)** −1.51 eV
- **(C)** −0.85 eV
- **(D)** −0.54 eV

20. If the velocity of an electron in the first Bohr orbit of hydrogen is v, its velocity in the second orbit will be:

- **(A)** v
- **(B)** v/2
- **(C)** 2v
- **(D)** v/4

21. The energy of the electron in the second orbit of hydrogen is −3.4 eV. The energy of the electron in the second orbit of the Li^{2+} ion is:

- **(A)** −3.4 eV
- **(B)** −10.2 eV
- **(C)** −30.6 eV
- **(D)** −1.13 eV

22. The ratio of the radii of the 2nd and 3rd Bohr orbits of the hydrogen atom is:

(A) 4 : 9
(B) 9 : 4
(C) 2 : 3
(D) 3 : 2

23. The minimum energy required to ionise a hydrogen atom from its ground state is:

(A) 13.6 eV
(B) 3.4 eV
(C) 1.51 eV
(D) 10.2 eV

24. An electron transition between the 2nd and 3rd Bohr orbits of hydrogen gives rise to a spectral line belonging to which series?

(A) Lyman
(B) Balmer
(C) Paschen
(D) Brackett

25. If the radius of the 3rd Bohr orbit of the hydrogen atom is r, the radius of the 3rd orbit of the He^+ ion is:

(A) r
(B) r/2
(C) 2r
(D) r/4

26. As the principal quantum number n increases in the hydrogen atom, the energy gap between two consecutive Bohr orbits:

(A) increases
(B) decreases
(C) remains constant
(D) first increases, then decreases

E. Hydrogen Spectrum

27. Which spectral series of the hydrogen atom lies in the ultraviolet region?

(A) Balmer series
(B) Lyman series
(C) Paschen series
(D) Brackett series

28. The wavenumber of the first line of the Balmer series of hydrogen ($n = 3 \rightarrow 2$) is about 15,233 cm^{-1}. The wavenumber of the second line of the same series ($n = 4 \rightarrow 2$) is approximately:

(A) 15,233 cm^{-1}
(B) 20,565 cm^{-1}
(C) 12,186 cm^{-1}
(D) 24,373 cm^{-1}

29. Which of the following electronic transitions in the hydrogen atom releases radiation of the highest energy?

(A) n = 2 to n = 1
(B) n = 6 to n = 5
(C) n = 3 to n = 2
(D) n = 5 to n = 4

30. The Paschen series of the hydrogen spectrum arises from transitions terminating at:

(A) n = 1
(B) n = 2
(C) n = 3
(D) n = 4

31. If the shortest wavelength (series limit) of the Lyman series of hydrogen is λ, the shortest wavelength of the Lyman series of the He^+ ion is:

(A) λ
(B) λ/2
(C) λ/4
(D) 4λ

F. Dual Nature of Matter — de Broglie Wavelength

32. The de Broglie wavelength of a moving particle is inversely proportional to its:

(A) mass only
(B) velocity only
(C) momentum
(D) charge

33. An electron is accelerated through a potential difference of 100 V. Its de Broglie wavelength is approximately ($\lambda \approx 1.227/\sqrt{V}$ nm):

(A) 0.123 nm
(B) 1.23 nm
(C) 12.3 nm
(D) 0.0123 nm

34. A ball of mass 0.1 kg moves with a velocity of 10 m s^{-1}. Its de Broglie wavelength is approximately ($h = 6.63 \times 10^{-34}$ Js):

(A) 6.63×10^{-34} m
(B) 6.63×10^{-33} m
(C) 6.63×10^{-32} m
(D) 6.63×10^{-35} m

35. Particles A and B have equal kinetic energy, but the mass of A is four times the mass of B. The ratio of their de Broglie wavelengths, λ_A : λ_B, is:

(A) 1 : 2
(B) 1 : 4
(C) 2 : 1
(D) 4 : 1

G. Heisenberg's Uncertainty Principle

36. Heisenberg's uncertainty principle states that it is impossible to determine simultaneously, with absolute precision, the:

(A) mass and charge of an electron
(B) position and momentum of a microscopic particle
(C) energy and charge of a particle
(D) mass and velocity of a macroscopic object

37. If the uncertainty in the position of an electron is 100 pm, the minimum uncertainty in its momentum is approximately ($\Delta x \cdot \Delta p \geq h/4\pi$, $h = 6.63 \times 10^{-34}$ Js):

(A) 5.3×10^{-25} kg m s^{-1}
(B) 5.3×10^{-22} kg m s^{-1}
(C) 1.05×10^{-24} kg m s^{-1}
(D) 1.05×10^{-27} kg m s^{-1}

38. Heisenberg's uncertainty principle is not noticeable for everyday macroscopic objects mainly because:

(A) macroscopic objects have no wave nature at all
(B) Planck's constant h is extremely small compared with the mass and dimensions involved
(C) macroscopic objects do not obey the laws of physics
(D) the principle applies only to charged particles

39. If the uncertainties in the position and velocity of a particle of mass m are numerically equal, the minimum uncertainty in velocity is given by:

(A) $\sqrt{(h/4\pi m)}$
(B) $h/4\pi m$
(C) $\sqrt{(h/4\pi)}/m$
(D) $4\pi m/h$

H. Quantum Numbers

40. The four quantum numbers of the valence electron of a sodium atom (Z = 11, configuration …$3s^1$) are:

(A) n = 3, l = 0, m = 0, s = +1/2
(B) n = 3, l = 1, m = 0, s = +1/2
(C) n = 2, l = 0, m = 0, s = +1/2
(D) n = 3, l = 0, m = 1, s = +1/2

41. Which of the following sets of quantum numbers is not possible for an electron?

(A) n = 3, l = 2, m = −2, s = +1/2
(B) n = 2, l = 1, m = 2, s = −1/2
(C) n = 4, l = 0, m = 0, s = +1/2
(D) n = 3, l = 1, m = 1, s = −1/2

42. The maximum number of electrons that can be accommodated in a subshell with n = 4, l = 3 is:

(A) 6
(B) 10
(C) 14
(D) 2

43. The orbital angular momentum of an electron present in a 3p orbital is:

(A) $\sqrt{2}$ $(h/2\pi)$
(B) $\sqrt{6}$ $(h/2\pi)$
(C) 0
(D) $\sqrt{3}$ $(h/2\pi)$

44. Which quantum number determines the orientation of an orbital in space?

(A) principal quantum number (n)
(B) azimuthal quantum number (l)
(C) magnetic quantum number (m)
(D) spin quantum number (s)

I. Electronic Configuration

45. For three electrons occupying the 2p subshell, which arrangement obeys Hund's rule of maximum multiplicity?

- **(A)** all three electrons paired in one orbital
- **(B)** two electrons in one orbital and one orbital left empty
- **(C)** one electron in each of the three 2p orbitals, all with parallel spin
- **(D)** two electrons in one orbital with opposite spins, and one electron in a second orbital

46. The electronic configuration of chromium (Z = 24), which departs from the simple Aufbau filling order, is best explained by:

- **(A)** Pauli's exclusion principle alone
- **(B)** the extra stability associated with a half-filled 3d subshell
- **(C)** Heisenberg's uncertainty principle
- **(D)** Hund's rule alone

47. According to Pauli's exclusion principle, no two electrons in the same atom can have:

- **(A)** the same principal quantum number
- **(B)** the same value for all four quantum numbers
- **(C)** the same azimuthal quantum number
- **(D)** opposite spins while in the same orbital

48. The number of unpaired electrons in the ground-state configuration of the Fe^{2+} ion (Z = 26, configuration $[Ar]3d^6$) is:

- **(A)** 2
- **(B)** 4
- **(C)** 6
- **(D)** 0

J. Quantum Mechanical Model — Orbital Shapes & Nodes

49. The total number of nodes (radial + angular) present in a 4d orbital is:

- **(A)** 1
- **(B)** 2
- **(C)** 3
- **(D)** 4

50. The characteristic shape of a p-orbital is:

- **(A)** spherical
- **(B)** dumb-bell
- **(C)** double dumb-bell (cloverleaf)
- **(D)** spherical with a node

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	A	21	C	31	C	41	B
2	C	12	A	22	A	32	C	42	C
3	B	13	A	23	A	33	A	43	A
4	A	14	B	24	B	34	A	44	C
5	C	15	A	25	B	35	A	45	C
6	C	16	A	26	B	36	B	46	B
7	C	17	B	27	B	37	A	47	B
8	A	18	B	28	B	38	B	48	B
9	B	19	C	29	A	39	A	49	C
10	A	20	B	30	C	40	A	50	B